Day 1 – Session 1
Introduction to VLSI and Analog Mixed SIGNAL Design

Session Speaker:

Dr. Cyril Prasanna Raj P.

Professor – Cambridge Institute of Technology

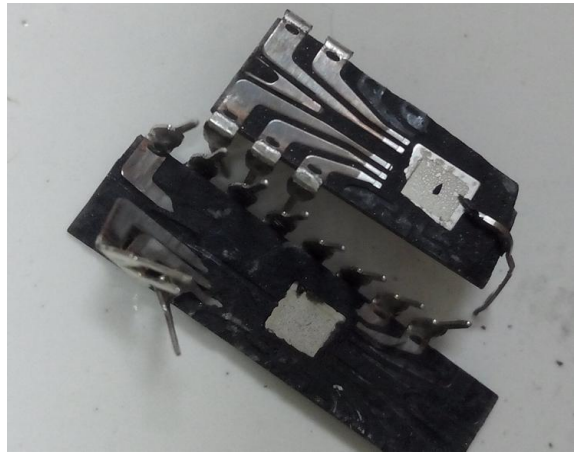
Director – Cambrian Consultancy Center and Industrial Research

cyril@cambridg.edu.in, cyrilyahoo@gmail.com

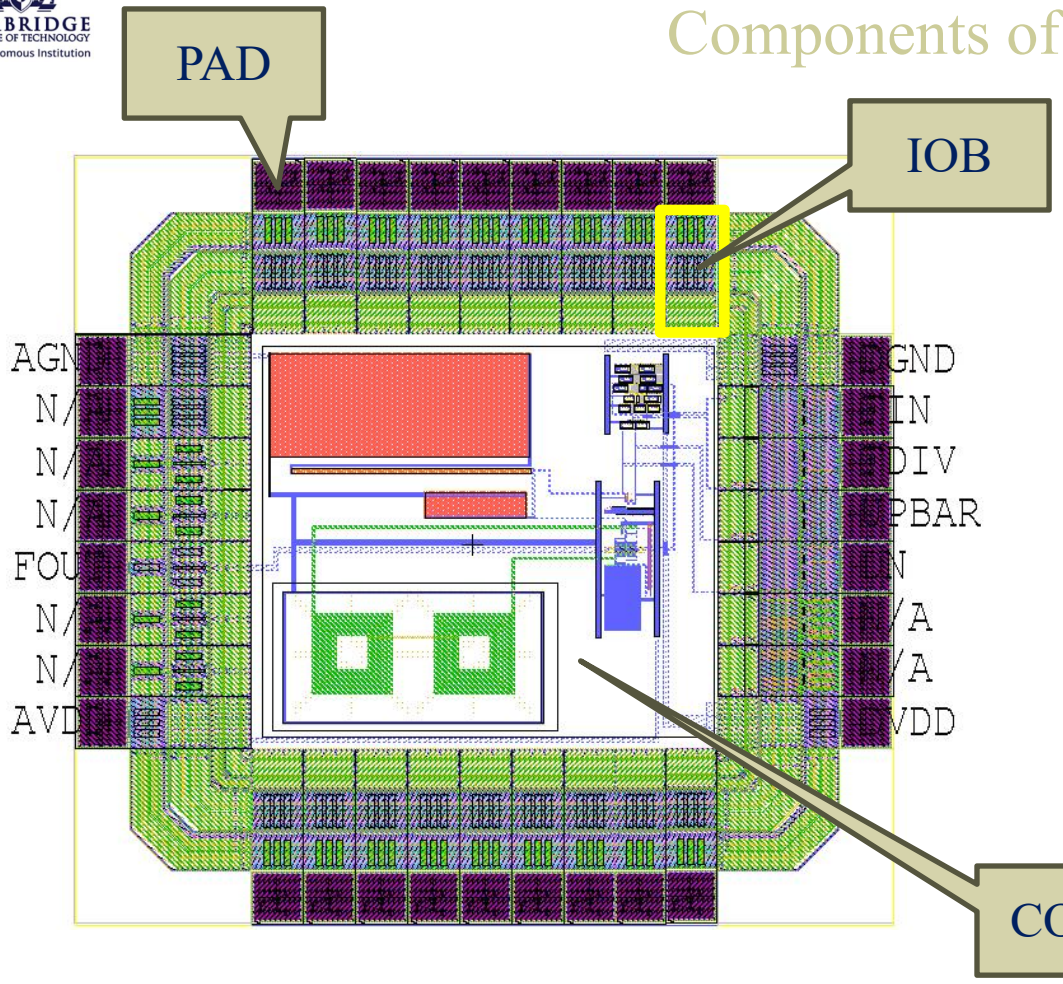
Answer these three questions

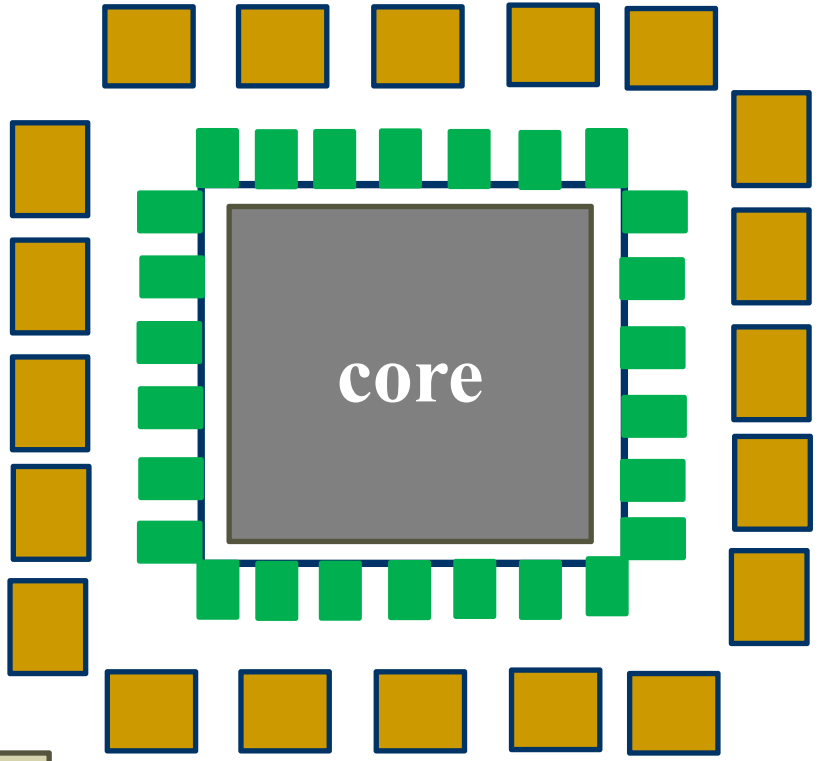
1. Integrated circuit is black (gray) in color ?
   A. Yes or No
2. Integrated circuit has pins on two sides (or four sides) ?
   A. Yes or No
3. The component in the picture is called integrated circuit ?
   A. Yes or No

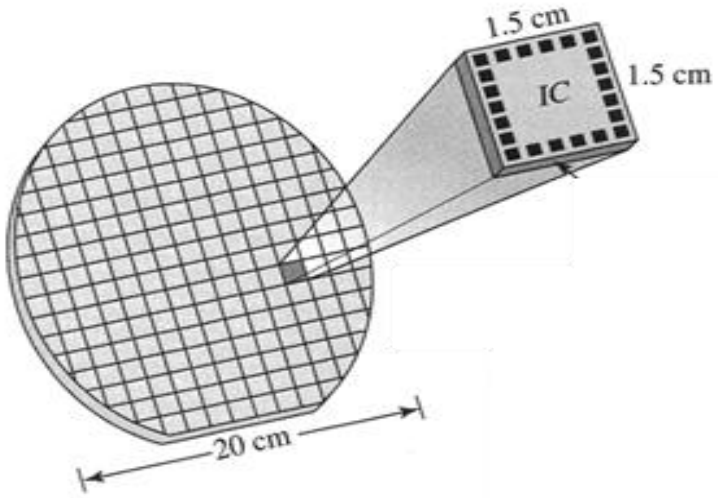




Components of Full Chip

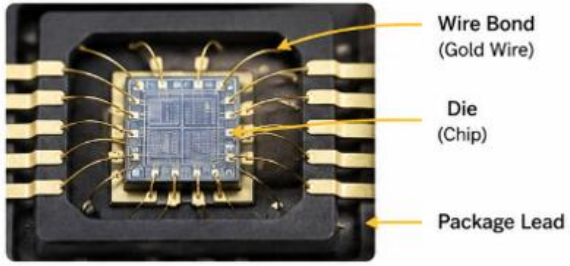






WIRE BOND

- Wire bonding uses fine metal wires to connect the bond pads on the semiconductor die to the package leads.

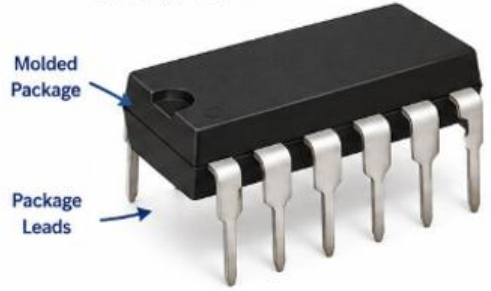


Functions

- Provide electrical connection between die and leads
- Enable signal and power transmission
- Accommodate thermal and mechanical stress

PACKAGE LEAD

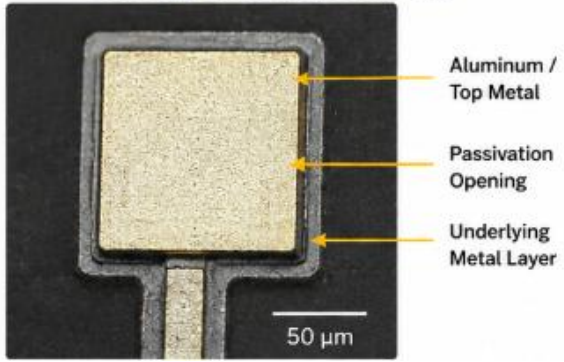
- Leads are the metal terminals of a package that connect the internal circuitry to the external PCB.



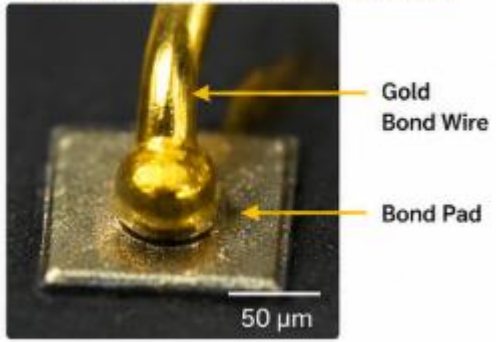
Functions

- Provide mechanical support
- Provide electrical connection to PCB
- Carry signal and power to/from the device

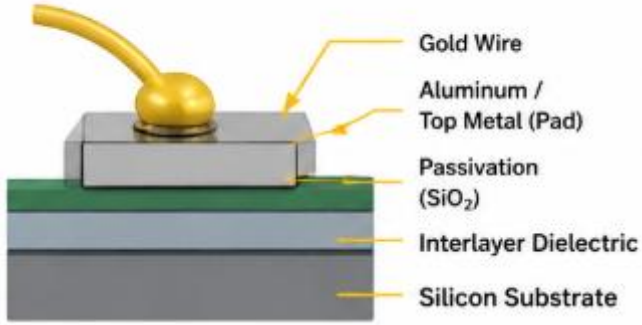
Close-up View of a Bond Pad



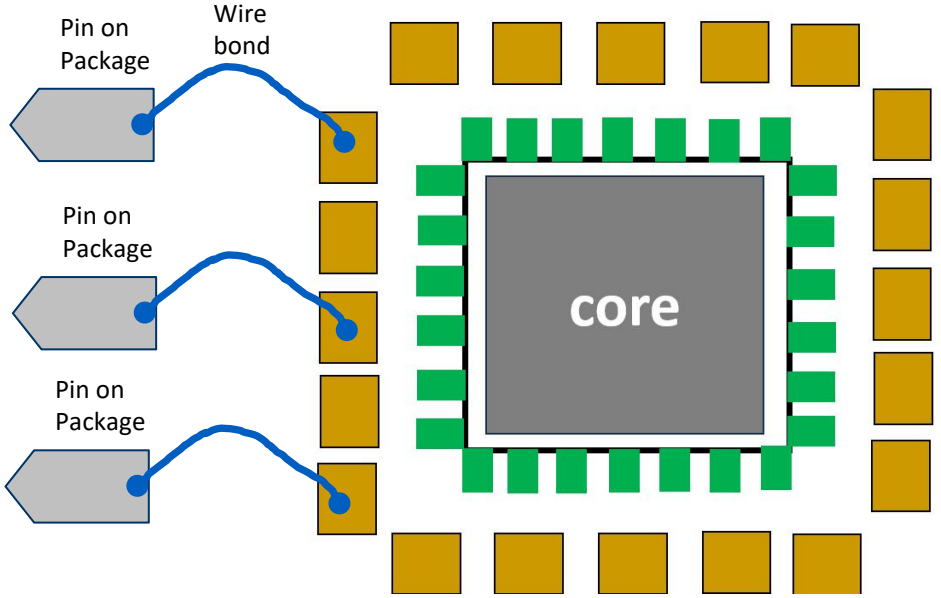
Bond Wire on Bond Pad (Actual)

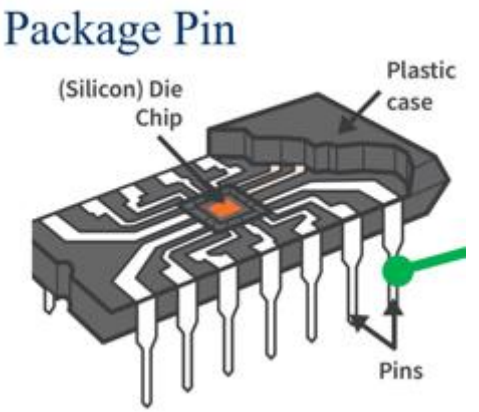


Cross-sectional View



VLSI Design



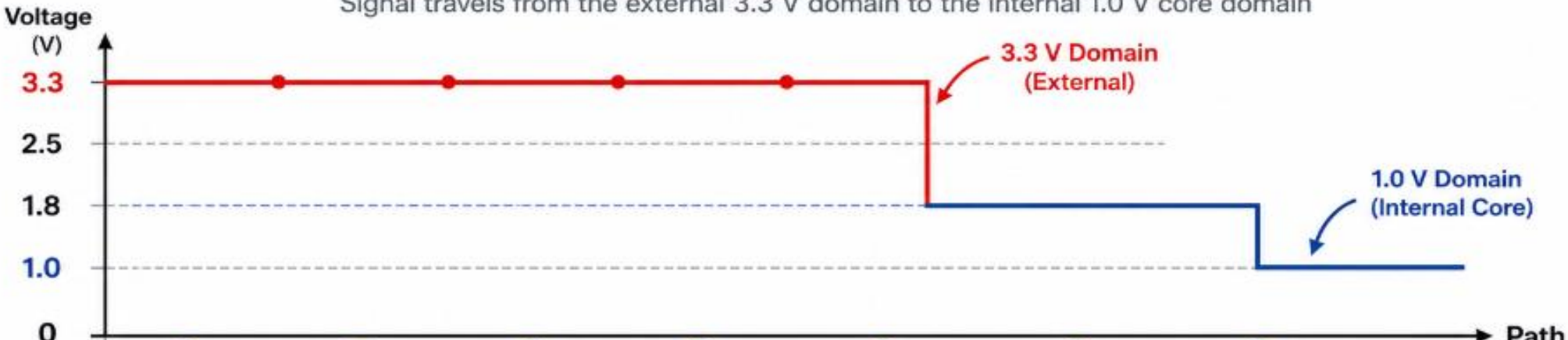


Parameter	Typical Value
Voltage	1.8 V, 2.5 V, **3.3 V**, sometimes 5 V tolerant
Current	2–32 mA per pin
Output Drive	4 mA, **8 mA**, 12 mA, 16 mA, 24 mA
Frequency	DC to several hundred MHz
Power	**26.4mW**

End-to-End Signal Transformation

VOLTAGE TRANSITION ACROSS THE SIGNAL PATH

Signal travels from the external 3.3 V domain to the internal 1.0 V core domain



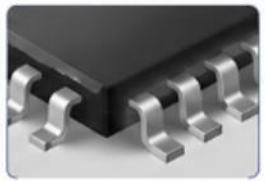
1 PCB



3.3 V | ~8 mA

Carries signals from connectors or other external components

2 Package Pin (Lead / Ball)



3.3 V | ~8 mA

Provides electrical and mechanical connection between PCB and package

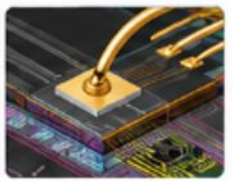
3 Wire Bond



≈3.3 V | ~8 mA

Gold/Cu wire connects package pin to bond pad on the silicon die

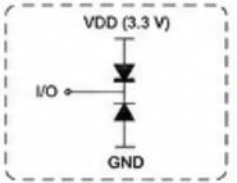
4 Bond Pad (on Die)



≈3.3 V | ~8 mA

Metal landing area on die for wire bond connection (signal entry point)

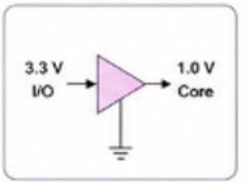
5 ESD Protection Circuit



3.3 V (normal)

Protects internal circuits from ESD / EOS events by clamping to VDD or GND

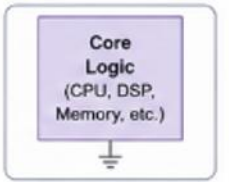
6 I/O Buffer (IOB) Level Shifter



3.3 V → 1.8 V → 1.0 V

Performs level translation, drive strength control, and signal conditioning

7 Core Logic (1.0 V Domain)

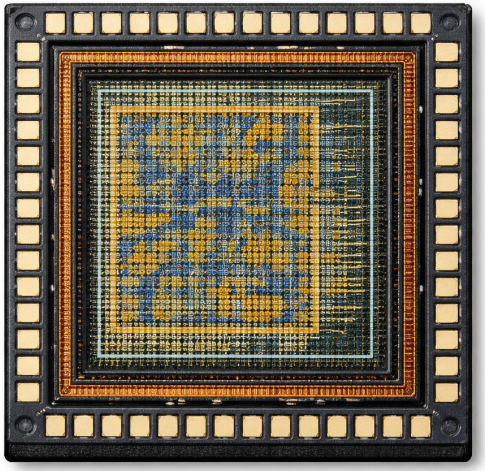


1.0 V

Low-voltage circuitry performs computation and data processing

Purpose of a CHIP

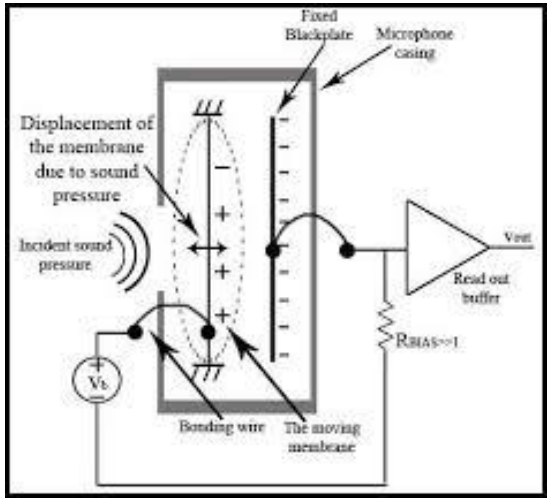




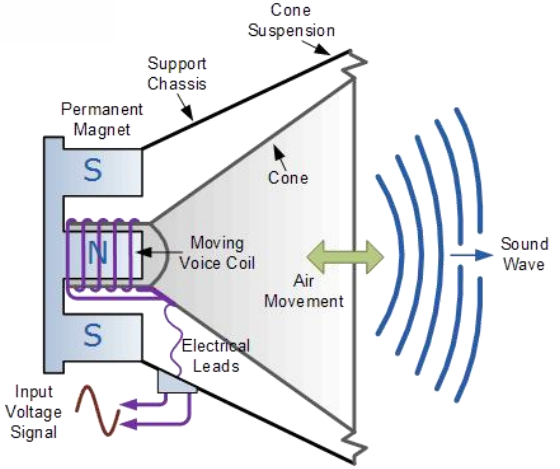


Sensor

Microphone



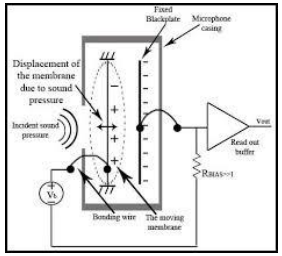
Speaker



Actuator

CHIP Functionality Not Limited ?

Sensor



PGA

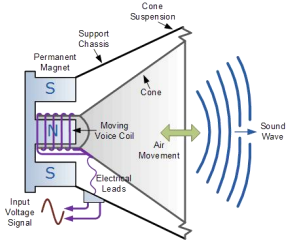
ADC

Digital

DAC

Driver/ filter

Actuator



Digital

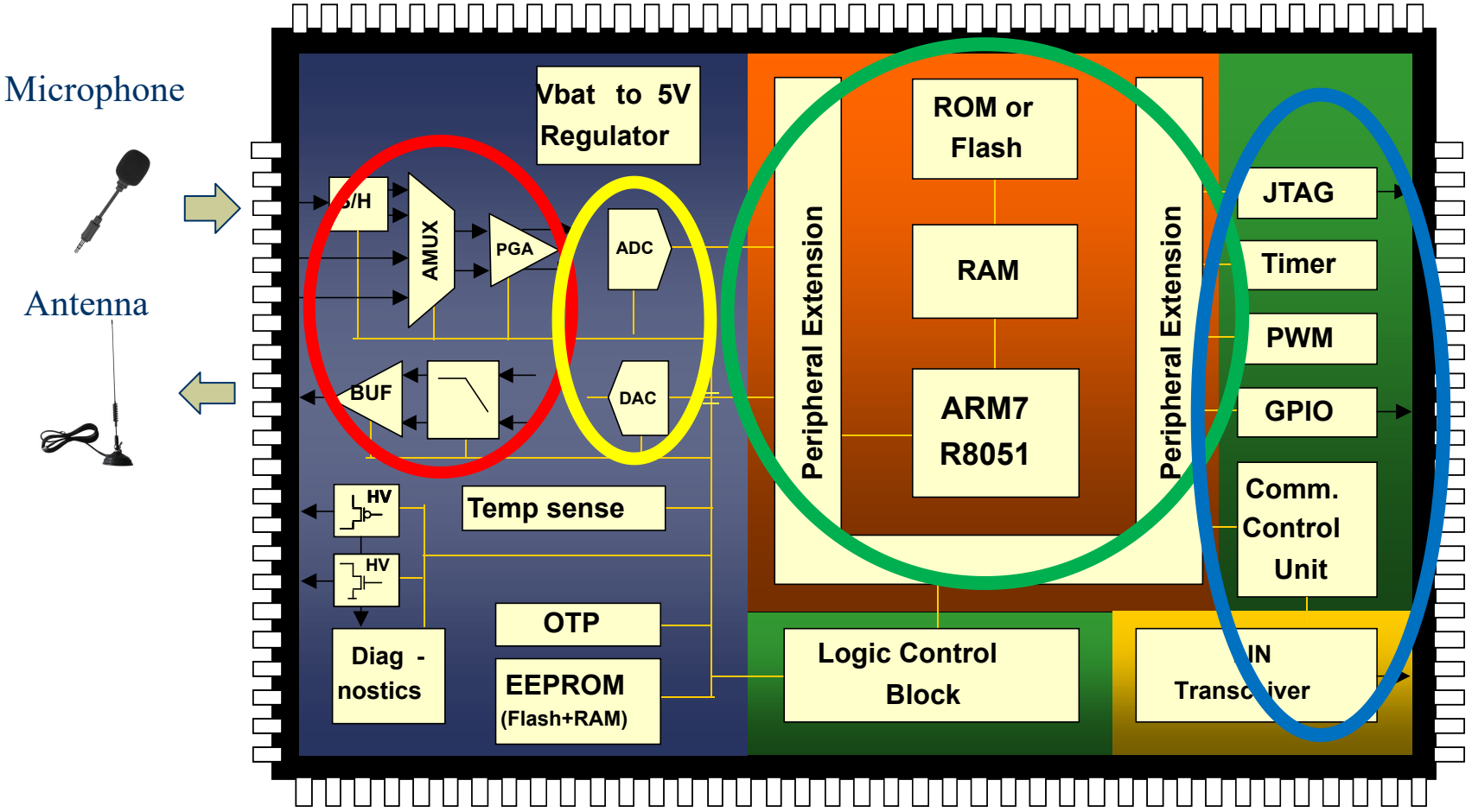
ADC

DAC

Driver/ filter

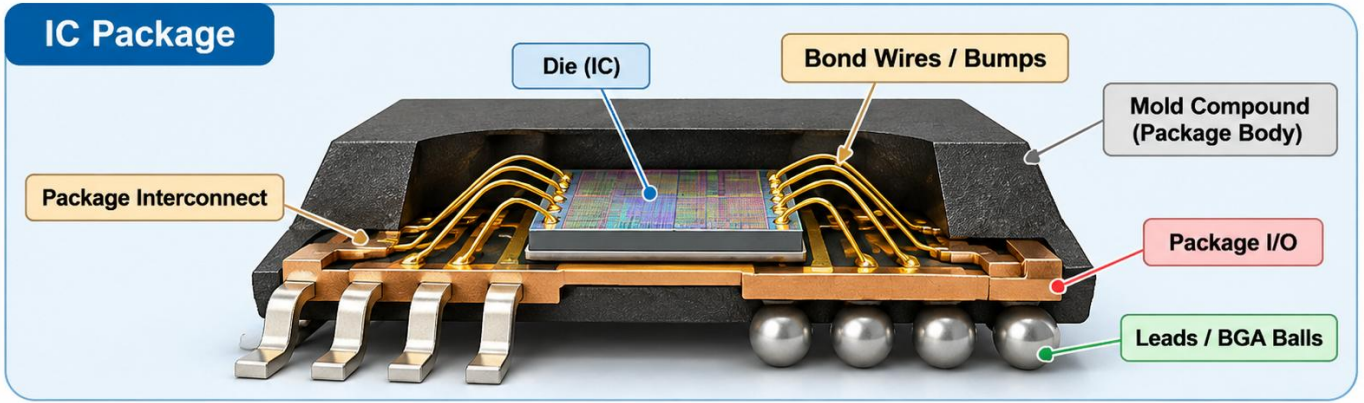
PGA

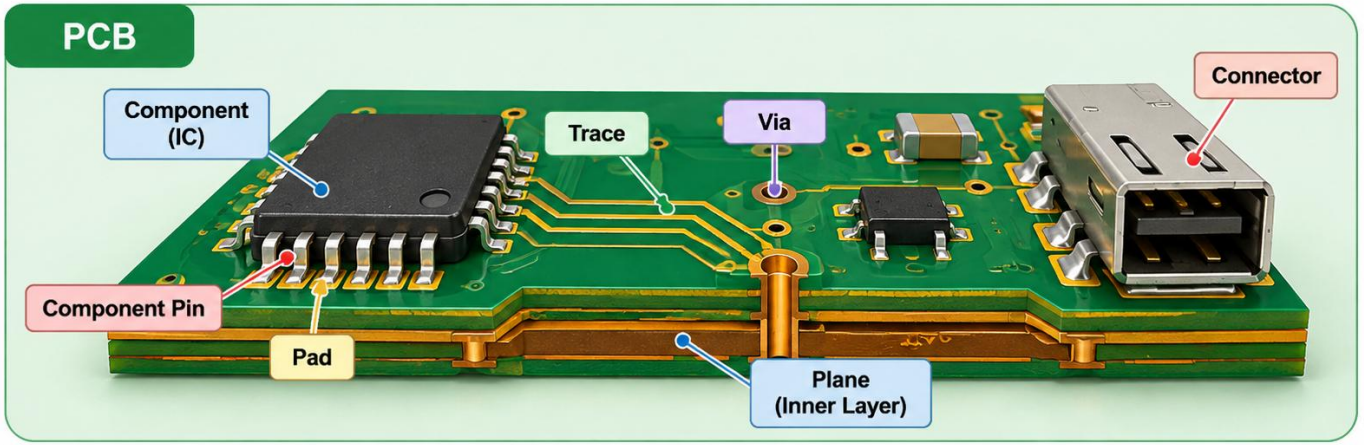
System on CHIP (SOC) – Internal Architecture



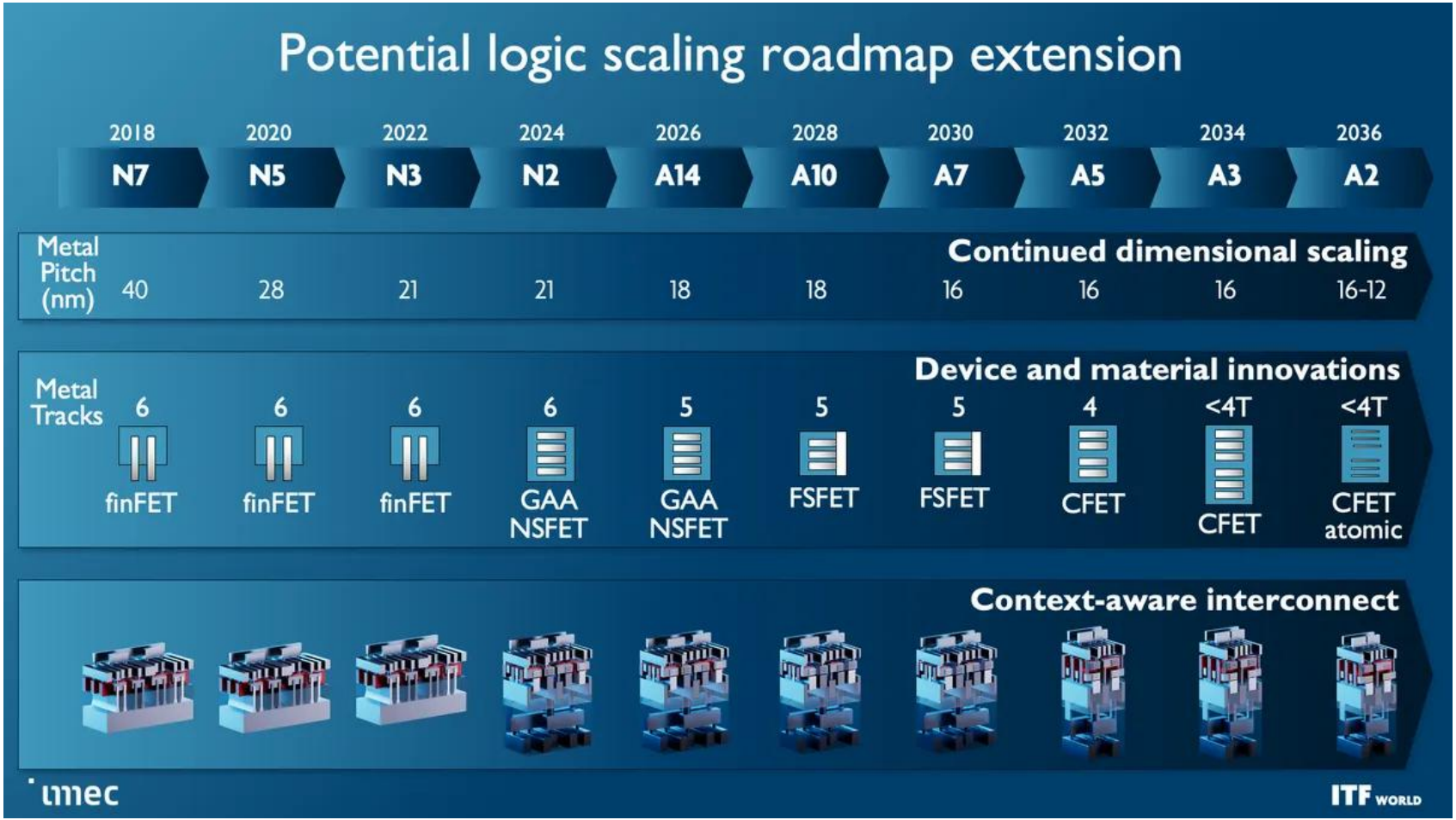
Three major components in chip design

- The die or chip
  - Core
  - IO blocks
  - Pads
- Package
  - Package Interconnect
  - Package I/O
  - Bond Wires / Bumps
  - Leads / BGA Balls
- PCB
  - Component/ component Pin
  - Pad
  - Trace
  - Via/ Plane
  - Connector





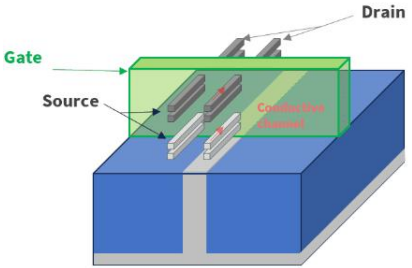
Transistor Technology



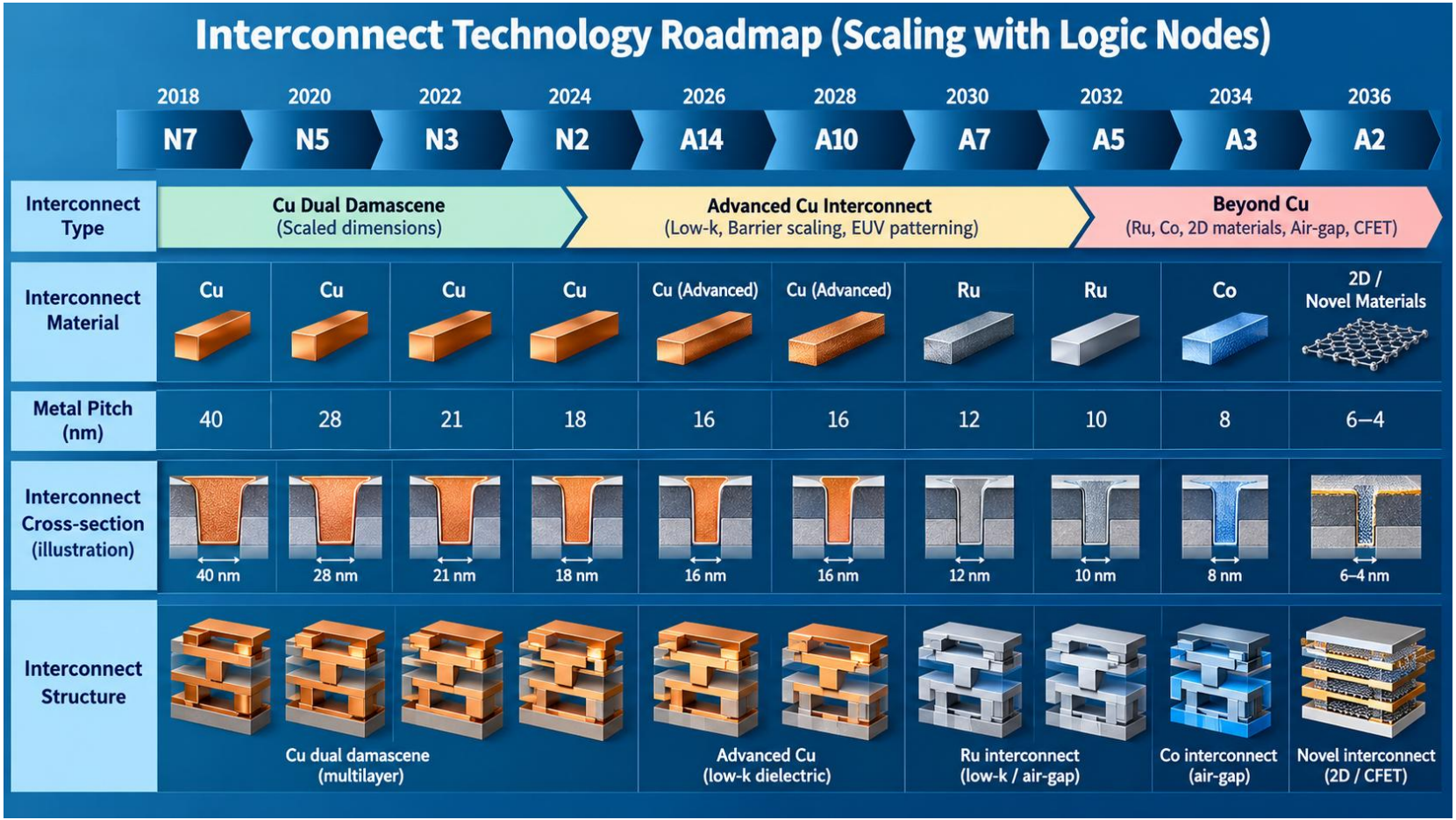
Floating Source Field-Effect Transistor

Nanosheet Field-Effect Transistor

Complementary Field-Effect Transistor



Interconnect Technology Road Map



Packaging

IC PACKAGING TYPES SUMMARY		
Common Types of IC Packaging	Advanced IC Packaging Types	Emerging IC Packaging Types/strong
• Dual InLine Packaging (DIP) • Small Outline Package (SOP) • Quad Flat Package (QFP) • Ball Grid Array (BGA)	• Chip-scale Packages (CSP) • System-in-Package (SiP) • Multi-Chip Modules (MCM) • 3D Packaging Techniques	• Fan-Out Wafer-Level Packaging (FOWLP) • Wafer-Level Chip-Scale Packaging (WLCSP) • Chiplet-based Packaging

small-outline transistor package

The most common IC package types include-

1.Dual In-Line Package (DIP)
2.Small Outline Package (SOP)
3.Thin Small Outline Package (TSOP)
4.Quad Flat Package (QFP)
5.Quad Flat Package-Extended (QFP-EP)
6.Quad Flat No-leads (QFN)
7.Ball Grid Array (BGA)
8.Micro Ball Grid Array (μBGA)
9.Ceramic Ball Grid Array (CBGA)
10.Plastic Ball Grid Array (PBGA)
11.SOT-223
12.SOT-323
13.SOT-363
14.SOT-523

1.Fine-Pitch Ball Grid Array (FBGA)
2.Chip Scale Package (CSP)
3.Dual Flat No-leads (DFN)
4.Plastic Leaded Chip Carrier (PLCC)
5.Small Outline Transistor (SOT)
6.Small Outline Diode (SOD)
7.Small Outline J-Lead (SOJ)
8.Dual In-line Memory Module (DIMM)
9.Single In-line Memory Module (SIMM)
10.Quad Flat Package No-lead (QFN-ML)
11.Ultra-Thin Quad Flat No-lead (UQFN)
12.SOT-23

IC Package Technology Roadmap (Scaling with Logic Nodes)

	2018 N7	2020 N5	2022 N3	2024 N2	2026 A14	2028 A10	2030 A7	2032 A5	2034 A3	2036 A2
Package Generation	Traditional Packages			Advanced Single-Die Packages		High-Density / Heterogeneous Integration		3D / Chiplet / System-in-Package		
Representative Package Types	DIP	SOIC	QFP	QFN	BGA	CSP	Flip-Chip	2.5D (Interposer)	3D (TSV)	Chiplet / Heterogeneous
Cross-section (illustration)	Wire bond (Through-hole)	Wire bond (Lead frame)	Wire bond (Lead frame)	Wire bond (Lead frame, no leads)	Wire bond (BGA balls)	Thin package (BGA/CSP)	Flip-chip (Solder/Cu bumps)	Multi-die + Interposer (Silicon/Organic)	3D stack (TSV)	Chiplets + Heterogeneous Integration (2.5D/3D, HBM, etc.)
I/O Density (relative)	~10^1	~10^1–10^2	~10^2	~10^2	~10^2–10^3	~10^3	~10^3–10^4	~10^4	>10^4	>10^4
	Low → High									
Package Size / Thickness (trend)	> 10 mm	~ 5–10 mm	~ 5–10 mm	~ 4–6 mm	~ 4–6 mm	~ 2–4 mm	~ 1–2 mm	~ 1–3 mm	~ 1–3 mm	~ 1–3 mm
	Larger / Thicker → Smaller / Thinner (Higher integration)									
Key Drivers / Use Cases	• Low cost • Through-hole assembly • Consumer electronics	• Surface mount • Higher pin count • Mobile, IoT	• Higher I/O count • Better electrical performance	• Smaller form factor • Consumer, mobile	• Higher I/O density • Mainstream high-performance SoCs	• Ultra-small size • Mobile, wearable, IoT	• Higher performance • Lower parasitics • HPC, AI accelerators	• Multi-die integration • Large bandwidth (e.g., HBM) • AI/HPC	• 3D stacking (TSV) • Higher bandwidth • Memory stacking	• Chiplet-based systems • Heterogeneous integration (logic, memory, RF, photonics) • AI, HPC, automotive

PCB Design Technology Roadmap (Next 10 Years)

Year	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034
Market / System Driver	Higher Performance and Higher Bandwidth		AI / HPC, 5G/6G, Automotive, Edge Computing			Lower Loss, Higher Integration, Heterogeneous Systems			System-level Integration (3D, Chiplets, Photonics, RF)	
Representative PCB Types	Standard Multilayer PCB	High-Speed Digital PCB	High-Frequency (RF/mmWave) PCB	HDI PCB	Server / AI Accelerator PCB	Automotive / High-Reliability PCB	Embedded Component PCB	2.5D/3D Integration PCB	Chiplet / Heterogeneous System PCB	Next-Gen (RF + Optical + Power)
Number of Layers (trend)	4 – 8 layers	6 – 12 layers	8 – 16 layers	12 – 20 layers	16 – 24 layers	20 – 32 layers	24 – 40 layers	> 40 layers	> 50 layers (3D / embedded)	> 50 layers (interposer-like)
Line Width / Spacing (trend)	≥ 100 µm	50 – 100 µm	30 – 50 µm	20 – 30 µm	10 – 20 µm	5 – 10 µm	3 – 5 µm	2 – 3 µm	≤ 2 µm	≤ 1 µm (advanced processes)
Via Technology (trend)	Through-hole (TH)	PTH (Plated Through Hole)	Blind Via	Buried Via	Stacked Via	Microvia (Laser drilled)	Filled / Stacked Microvia	Ultra-fine Microvia	3D Via (Vertical interconnect)	Hybrid Via (Electrical + Optical)
Material (Dielectric)	Standard FR-4	Low-loss FR-4	High-Tg FR-4	Low-loss / Very Low-loss (e.g. Rogers)	Ultra-low loss (e.g. PTFE, LCP)	Hybrid materials (Organic + Ceramic)	Glass core / Build-up	Advanced composites (High thermal)	Embedded dielectrics (for 3D)	Novel materials (Optical / Terahertz)
Design Tools & Methods	2D layout / DRC	Multi-layer routing	High-speed SI/PI analysis	3D field solver	Co-design (PCB + Package)	Thermal / Mechanical co-simulation	AI-assisted design (optimization)	Digital twin (system-level)	3D system design (PCB + Package + Chiplets)	Co-design with Photonic / RF
Key Applications	• Consumer electronics • Industrial • IoT	• Smartphones • Laptops • Networking	• 5G infrastructure • Automotive • ADAS	• High-speed computing • Consumer AI devices • Wearables	• AI servers • Data centers • HPC	• Autonomous vehicles • Industrial automation • Aerospace / Defense	• High-frequency RF • Automotive radar • Satellite • Edge AI	• 3D integration • Embedded components • High-bandwidth memory	• Chiplets • Heterogeneous systems • AI accelerators • Advanced computing	• RF + Optical • Next-gen communication • Quantum / Terahertz • AI / HPC systems
Key Trends	Higher Density and Higher Speed		Lower Loss and Higher Reliability			Heterogeneous Integration (2.5D / 3D)			System-Level Convergence (Electrical + RF + Optical + Power)	

Worksop Overview

First hand experience on Mixed Signal IC Design

14-bit SAR ADC Design and Tapeout Readiness

Day 1
AMS flow reference design

Day 2
SAR ADC design and subsystem integration

Day 3
SAR ADC verification

Day 4
SAR ADC layout design and optimization

Day 5
Physical verification and submitting design files for chip tapeout

Invited Lecture Sessions

Analog Design for Automotive & Aerospace,
Muhammed Shihab T A
Safran Electronics & Defense

Transceiver design for high-speed wireless systems,
Dr. G. S. Javed
Analog IP Designer

Analog IP Design
Rajkumar Upadhyay – HCL

Analog Layout & IO Cell Design Techniques,
Prakash Reddy, Quantumsilicons Pvt. Ltd.

Chip tapeout design challenges: best practices and way forward
Venkata K. Reddy – CDAC

Design of Analog Circuits for Biomedical Sensors
Dr. Radhakrishnan – PoP, CIT

EDA ecosystem

Function	Cadence	Synopsys	Siemens EDA
Custom/Analog Design	Virtuoso Studio	Custom Compiler	Custom IC Design
Analog Simulation	Spectre	PrimeSim	Solido / AFS / Eldo
AMS Simulation	Spectre AMS Designer	VCS AMS / PrimeSim	Symphony
Digital Simulation	Xcelium	VCS	Questa
Digital Synthesis	Genus	Design Compiler	Oasys-RTL
Digital P&R	Innovus	IC Compiler II	Aprisa
Analog Layout	Virtuoso	Custom Compiler	Custom IC / L-Edit
Parasitic Extraction	Quantus	StarRC	Calibre xACT
Physical Verification	Pegasus / PVS	IC Validator	Calibre
Reliability	Legato / Spectre	PrimeSim Reliability	Solido / Calibre PERC
EM/IR	Voltus-Fi	PrimeSim / related solutions	mPower
RF/EM	EMX / Clarity	Synopsys + Ansys integration	Siemens RF/EM solutions
Final physical sign-off	Pegasus/PVS	ICV	Calibre

Mixed Signal Design Flow in Cadence Environment

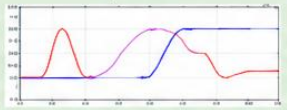
Mixed Signal System Level Design
(Architecture, modeling and verification)

cadence Virtuoso AMS Designer | Spectre | Verilog-AMS | SystemVerilog | MATLAB/Simulink co-simulation

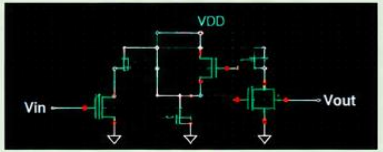
cadence MATLAB/Simulink™

Analog Cells / Blocks

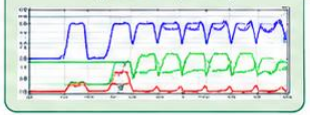
Electrical Simulation
cadence Spectre®



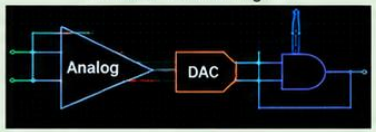
Analog Schematic Design
cadence Virtuoso® Schematic Editor



Mixed Signal Simulation
cadence Spectre® (AMS)



Mixed Signal Schematic Design
cadence Virtuoso® AMS Designer



Verification
cadence PVS® / Xcelium® / Spectre® (DRC/LVS/Signoff)

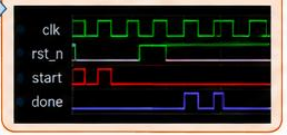
- DRC
- LVS
- ERC
- AMS simulation (post-layout)
- Signoff

Digital Cells / Blocks

High Level Description (HDL - RTL)
cadence Jasper® / Xcelium® (simulation)

```
module control (
  input clk,
  input rst_n,
  input start,
  output done
);
  ...
endmodule
```

Logic Simulation
cadence Xcelium®

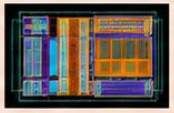


Digital Netlists
- Verilog / SystemVerilog
- Encrypted IP
- Synthesized netlists

Gate-Level Synthesis
cadence Genus™ Synthesis Solution

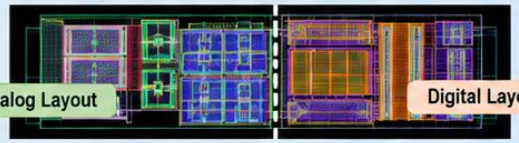
synthesis constraints (SDC)
timing reports

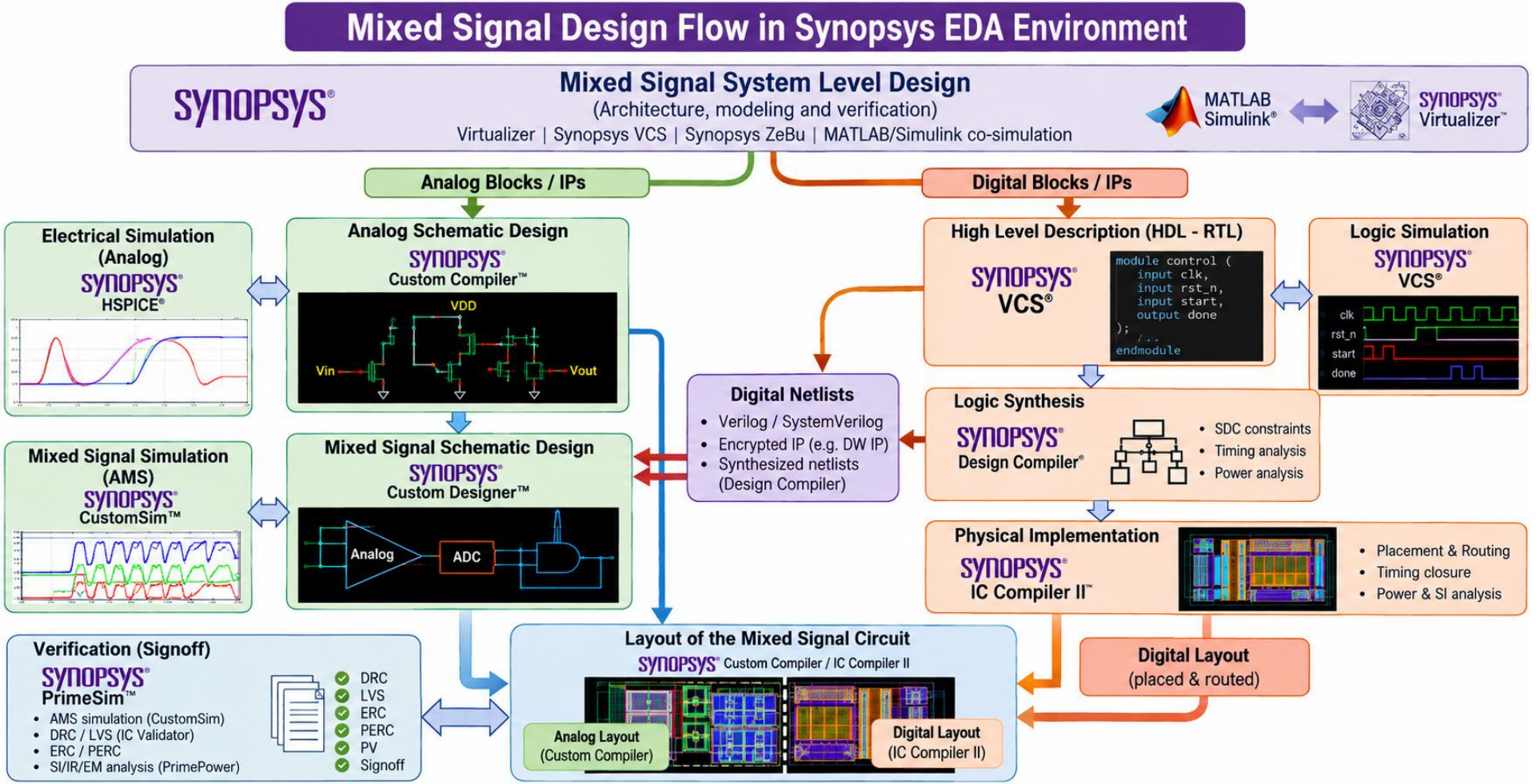
Automatic Place and Route
cadence Innovus™ Implementation System



Digital Layout
(placed & routed)

Layout of the Mixed Signal Circuit




Mixed Signal Design Flow in Synopsys EDA Environment
SYNOPSYS
Mixed Signal System Level Design
(Architecture, modeling and verification)
Virtualizer | Synopsys VCS | Synopsys ZeBu | MATLAB/Simulink co-simulation
MATLAB Simulink
Synopsys Virtualizer
Analog Blocks / IPs
Digital Blocks / IPs
Electrical Simulation (Analog)
SYNOPSYS HSPICE
Analog Schematic Design
SYNOPSYS Custom Compiler
VDD
Vin
Vout
High Level Description (HDL - RTL)
SYNOPSYS VCS
module control (
input clk,
input rst_n,
input start,
output done
);
endmodule
Logic Simulation
SYNOPSYS VCS
clk
rst_n
start
done
Digital Netlists
Verilog / SystemVerilog
Encrypted IP (e.g. DW IP)
Synthesized netlists (Design Compiler)
Logic Synthesis
SYNOPSYS Design Compiler
SDC constraints
Timing analysis
Power analysis
Mixed Signal Simulation (AMS)
SYNOPSYS CustomSim
Mixed Signal Schematic Design
SYNOPSYS Custom Designer
Analog
ADC
Physical Implementation
SYNOPSYS IC Compiler II
Placement & Routing
Timing closure
Power & SI analysis
Verification (Signoff)
SYNOPSYS PrimeSim
AMS simulation (CustomSim)
DRC / LVS (IC Validator)
ERC / PERC
SI/IR/EM analysis (PrimePower)
DRC
LVS
ERC
PERC
PV
Signoff
Layout of the Mixed Signal Circuit
SYNOPSYS Custom Compiler / IC Compiler II
Analog Layout (Custom Compiler)
Digital Layout (IC Compiler II)
Digital Layout (placed & routed)

Industry Standard Mixed Signal (AMS) Design Flow

1. System Specification
Requirements, architecture, performance targets

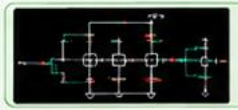
MATLAB/Simulink | Python | SystemC

2. Behavioral Modeling
Algorithm development, high-level modeling

Verilog-A / Verilog-AMS
SystemVerilog, MATLAB

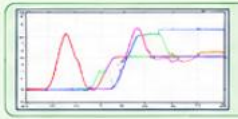


3. Analog / Custom Flow



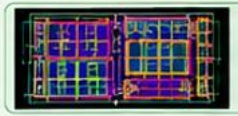
Analog Schematic Design
Circuit design and verification

Virtuoso / Custom Compiler / Custom IC



Analog Simulation
Circuit simulation (DC, AC, Transient, PVT, Monte Carlo, Noise)

Spectre / PrimeSim / Solido



Analog Layout
Custom layout (mainly manual)

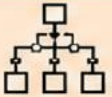
Virtuoso / Custom Compiler / L-Edit

4. Digital Flow



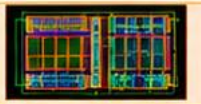
RTL Design & Verification
RTL coding, functional verification

Xcelium / VCS / Questa



Logic Synthesis
Synthesis, timing, power analysis

Genus / Design Compiler / Oasys-RTL



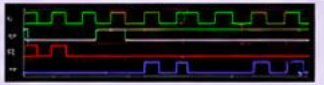
Place and Route
Digital implementation

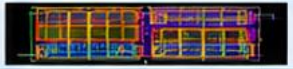
Innovus / IC Compiler II / Aprisa



5. AMS Co-simulation and Verification
Analog + Digital interaction, mixed-signal verification

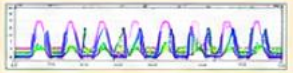
Spectre AMS / VCS-AMS / Symphony





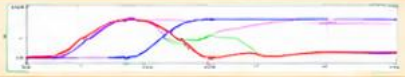
6. Parasitic Extraction (PEX)
Extract parasitics (RC, coupling)

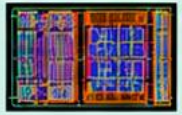
Quantus / StarRC / Calibre xACT



7. Post-layout Simulation
Include extracted parasitics, PVT, Monte Carlo

Spectre / PrimeSim / Solido





8. Physical Verification
Design rule check, layout versus schematic, ERC, antenna, etc.

Pegasus / ICV / Calibre

- DRC
- LVS
- ERC
- Antenna



9. Reliability & Sign-off
EM/IR, ESD, latch-up, aging, DFM, PERC, final sign-off

Legato / PrimeSim / Solido / Calibre PERC

- EM/IR
- Reliability
- DFM (PERC)
- Yield

10. Tape-out
GDSII release to foundry

